Tutorial 9. Gaussian Splatting and NeRF

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Interactive lab: sosekie.github.io/Unibe-CVG-CV-Interactive-Labs/tutorial-09/
Source code: github.com/Sosekie/Unibe-CVG-CV-Interactive-Labs

Learning objectives

- Relate Gaussian shape to covariance, PCA, and function approximation.
- Derive projection, compositing, and fitting behaviour from the underlying models.
- Compare Gaussian Splatting and NeRF and evaluate reconstruction evidence.

Key concepts. Least squares, covariance, gradient descent, alpha compositing, multi-view geometry, volume rendering, and neural radiance fields.

Explain your reasoning and show relevant derivations. The interactive lab provides experiments for each exercise.

1. Approximate a function in 1D

Objective. Distinguish amplitude fitting, basis geometry, and identifiability.

1.1 Identifiability - Linear Fitting

You fit two 1D Gaussian basis functions with exactly the same mean and width. Only their amplitudes are free. The target samples are fixed.

(a) What can least squares determine about the two amplitudes? Justify your answer.
(b) Would adding a penalty $\lambda\|a\|^2$ make the amplitudes identifiable from the observations?

1.2 Model Assumptions - Signed Weights

After an unconstrained amplitude fit, a nonnegative target is approximated using both positive and negative Gaussian coefficients. Small regions of the approximation fall below zero.

(a) Which interpretation is consistent with the model used in this lab? Justify your answer.
(b) Why can a low sample error coexist with large, unstable positive and negative coefficients?

Interactive experiment 1.

1. Fit the smooth-bump target with N = 2 and N = 8. Record sample RMSE and basis rank.
2. Make two kernels identical in mean and width. Predict the rank and possible amplitude solutions before fitting.
3. Try the signed-oscillation target. Inspect the individual contributions and explain negative coefficients.

Record settings, numerical results, and observations. Open the corresponding experiment.

2. Covariance and a 2D input domain

Objective. Read Gaussian geometry from covariance eigenpairs and distinguish scale from normalization.

2.1 Covariance - Principal Axes

A 2D Gaussian has $\Sigma = \begin{bmatrix} 0.20 & 0.16 \\ 0.16 & 0.20 \end{bmatrix}$. The diagonal entries are equal.

(a) What shape does this covariance describe? Justify your answer.
(b) What happens to the usefulness of a rotation gradient as the two principal scales become equal?

2.2 Gaussian Behavior - Normalization

Consider one unnormalized 2D kernel $a \exp(-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu))$ with positive amplitude a > 0. Double both principal scales while keeping a fixed.

(a) What happens to its peak value and its integral over the whole plane? Justify your answer.
(b) Does the fourfold integral increase imply fourfold image brightness after projecting and alpha-compositing a 3D splat?

Interactive experiment 2.

1. Set one Gaussian's scales to 0.60 and 0.20 and its rotation to 45°. Inspect its covariance.
2. Make it isotropic and rotate again. Explain which quantities stop changing and why.
3. Isolate one contribution and double both scales at fixed amplitude. Predict the peak and full-plane integral; account for the finite plot and automatic display scaling.

Record settings, numerical results, and observations. Open the corresponding experiment.

3. Fit a 3D scalar function

Objective. Separate approximation capacity, initialization, and optimization error.

3.1 Optimization - Gradient Flow

The 3D model is $\hat{y}(\mathrm{x}) = a\phi(x; \mu, \Sigma)$, fitted by mean squared error to sampled target values. Consider the raw gradient at a = 0, with positive scales.

(a) Which parameters can receive a nonzero loss gradient at that instant? Justify your answer.
(b) Can Adam momentum move a mean when its current raw gradient is zero?

3.2 Approximation - Capacity Vs Optimization

One Gaussian fits the ring-shaped 3D target poorly after many updates. Its slices fill the hole or miss much of the ring.

(a) What can be concluded from the Gaussian model itself, without assuming the optimizer found a global minimum? Justify your answer.
(b) If the displayed volume RMSE is small, could a visually important part of the ring still be poorly fitted?

Interactive experiment 3.

1. Choose the Gaussian-bump target and N = 1. Set amplitude to zero, record geometry, and apply one update. Explain the raw-gradient effect.
2. Fit the ring target with N = 1 and N = 8 using comparable update budgets. Inspect all three residual slices.
3. Rotate the inspection view without editing the fit. Explain why the sampled scalar objective stays fixed.

Record settings, numerical results, and observations. Open the corresponding experiment.

4. Project, composite, and expose ambiguity

Objective. Relate visibility to gradients and distinguish rotation from translation.

4.1 Visibility - Photometric Gradients

At one pixel, a foreground splat has $\alpha_1 = 0.98$ and a rear splat has $\alpha_2 = 0.6$. Hold geometry and opacity fixed while differentiating the pixel colour with respect to the rear RGB.

(a) What is $\partial C/\partial c_2$ for each RGB channel, and what does it imply for fitting? Justify your answer.
(b) Would directly adding the two coloured kernels give the same learning behavior?

4.2 Multi-View Geometry - Ambiguity

For an isolated Gaussian and a fixed pinhole camera, multiply its camera-relative mean and all spatial scales by $k > 0$. Keep RGB and opacity fixed, and keep it in front of the near plane. Its locally projected footprint is unchanged.

(a) Which extra observation generally constrains this depth-scale ambiguity? Justify your answer.
(b) Why can three nearby training views provide less useful depth information than three spread-out views?

Interactive experiment 4.

1. Use three-splat overlap. Sample one pixel, vary foreground opacity, and follow the rear colour weights.
2. Keep one Gaussian. Record its projected axes, move it along the viewing ray with the provided scale change, and compare.
3. Change camera yaw to expose the 3D difference. Here yaw orbits the camera centre; explain why pure rotation at a fixed centre is different.

Record settings, numerical results, and observations. Open the corresponding experiment.

5. Recover a scene from images

Objective. Design controlled comparisons and distinguish image agreement from geometric evidence.

5.1 Experiment Design - Generalization

A fit from one selected camera has low training MSE but a much larger reserved-view MSE. You want to test whether limited view coverage is contributing to the failure.

(a) Which comparison provides the most relevant controlled evidence? Justify your answer.
(b) If you repeatedly choose budgets and stop times using frame 10, is it still an untouched final test?

5.2 Density Control - Model Changes

Consider one isolated coloured splat with opacity o. Replace it by two coincident splats with identical shape and colour, each with opacity $o' = 1 - \sqrt{1 - o}$. Let ϕ be their shared projected Gaussian value at a pixel.

(a) Does this opacity adjustment preserve the rendered image everywhere? Justify your answer.
(b) For an additive scalar field, would two coincident copies with amplitudes a/2 preserve the function?

Interactive experiment 5.

1. Fit one training view with 24 Gaussians for 100 updates. Record training and reserved-view MSE.
2. Restart with three spread-out views, using the same budget and update count. Compare both errors and residual structure.
3. Record error and count, apply density refinement once, then fit 25 further updates. Separate the immediate model change from subsequent learning.

Record settings, numerical results, and observations. Open the corresponding experiment.

6. NeRF: a neural field along camera rays

Objective. Distinguish density, segment opacity, quadrature, and neural-field capacity.

6.1 Volume Rendering - Optical Thickness

A ray crosses a homogeneous slab of length $L = 1$ with density $\sigma = 2$ and constant RGB. You approximate it with N equal intervals of length $\delta = L/N$, using $\alpha_i = 1 - \exp(-\sigma\delta)$.

(a) What happens to total transmittance if you change N from 10 to 100? Justify your answer.
(b) If a thin high-density feature lies entirely between the sample points, does a low sampled density prove the ray is empty?

6.2 Neural Fields - Multi-View Consistency

In a classical NeRF-style model, density is $\sigma_\theta(x)$, while RGB is $c_\theta(x, d)$. A proposed variant also lets density depend freely on viewing direction: $\sigma_\theta(x, d)$.

(a) Why is position-only density a useful constraint for reconstructing a static scene? Justify your answer.
(b) Is increasing the number of samples per ray the NeRF equivalent of adding more Gaussian primitives?

Interactive experiment 6.

1. At yaw 90° and the centre ray, compare density with termination weights. Explain the weak contribution of hidden orange density.
2. Compare 8 and 64 ray samples with fixed network weights. Track interval length, alpha, and total transmittance.
3. Fit 50 updates and inspect the reserved 0° view. Compare errors and explain why additional ray samples do not add network parameters.

Record settings, numerical results, and observations. Open the corresponding experiment.

7. Transfer to the real captured Garden

Objective. Evaluate appearance and geometry claims at scene scale.

7.1 Appearance - Representation Choice

Suppose a glossy leaf changes brightness with viewing direction in the captured photographs, but a reconstruction represents each Gaussian with one fixed RGB value. Even with good spatial coverage, a highlight appears painted onto the surface.

(a) Which model change directly addresses this missing behavior? Justify your answer.
(b) Could a low photometric error prove that positions and covariances are correct when colour is very flexible?

7.2 Transfer - Rendering Vs Geometry

The garden looks sharp from several viewpoints. A colleague wants to treat the visible Gaussian ellipsoids as a reliable surface model for collision detection.

(a) What does the rendering evidence justify? Justify your answer.
(b) If rendering at twice the resolution reveals smaller floaters, has the underlying geometry become worse?

Interactive experiment 7.

1. Load the Garden when network and WebGL are available. It streams approximately 187 MB and renders Gaussian Splatting.
2. Inspect thin structures and viewpoints away from the comfortable viewing region. Record two artifacts and competing explanations.
3. Relate observations to coverage, representation, or appearance; identify evidence needed before using this renderer as collision geometry.

Record settings, numerical results, and observations. Open the corresponding experiment.